

Look What We've Got for You on This Month's DVD!

In this month's EFY Plus DVD, we deliver some free embedded design and robotics programming tools. In case you are the adventurous kind, we have also included two special tools for neural networks—Neural Network Designer and iqr Neural Network Simulator

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Kactus2 (with an article featured in this issue)

Kactus2 is the first graphical open source IP-XACT toolset to design embedded products. It has features for packetising IP for reuse and exchange, and can create electronic datasheets of your existing IPs to simplify integration. It can also import, export and check the integrity of IP libraries from any standard compatible IP vendor.

Additionally, Kactus2 can be used to draft block diagram blueprints for IP, system-on-chip and printed circuit boards (PCBs). The product is stored in an IP-XACT format. This tool is handy for creating FPGA-based MP-Soc products with hierarchical hardware designs and also system designs for mapping software to hardware.

Supported OS: Windows
Size: 25 MB

The iqr large-scale neural systems simulator (full article featured in this issue)

iqr is a simulation tool for large-scale neural networks. It introduces students to neuronal simulation and neuromorphic control. iqr provides a graphical interface for designing neuronal models as well as graphical online control of the simulations. It has predefined interfaces to robots, cameras and other

hardware, and presents a multilevel neuronal simulation environment.

Supported OS: Linux, MacOS X and Windows
Size: 12 MB

Neural Network Designer

NND consists of a Windows-based designer application and a library. It is a database management system for neural networks. The included library is optimised for handling any type of data-size by using advanced streaming and caching algorithms. It can be used to create different types of decision trees, random forests, n-grams, pattern-matches, conversational agents and all sorts of AI-related algorithms. It also features rich GUIs for pattern matchers, queries, mind maps, etc. The installer includes demo projects, a set of topics and online applications, which can be installed from within the designer.

Supported OS: Windows, Android
Size: 48 MB (Windows)

RapidSmith

RapidSmith is a research-based, open source FPGA CAD tool written in Java for modern Xilinx FPGAs. Based on XDL, its objective is to serve as a rapid prototyping platform for research ideas and algorithms relating to low-level FPGA CAD tools. The new version 0.5.2 also has preliminary support for Artix 7 and Zynq devices.



OS: Windows
Size: 345 MB

YARP

YARP or Yet Another Robot Platform is a C++ based robot control system available for Windows, Linux and MacOS. This tool supports building a robot control system as a collection of programs communicating in a peer-to-peer way. It comprises an extensible family of connection types (tcp, udp, multicast, local, MPI, mjpeg-over-http, XML/RPC, tcpros, etc).

YARP supports flexible interfacing with hardware devices.

Supported OS: Linux, MacOS X and Windows.
Size: 40 MB

Simulide

Simulide is an open source circuit simulator that can be quickly installed on Windows. It is an easy-to-use platform for circuit simulation. The tool allows the user to insert an Arduino Uno board, add diodes, resistors, voltage sources and various micro AVR